Marketing Performance Analysis

End-to-End Data Analysis Project

Python | pandas | matplotlib | seaborn

Data Cleaning | KPI Definition | EDA | Business Insights

1. Project Overview

This project covers a full end-to-end marketing analysis on a synthetic dataset containing ~3,000 rows of marketing performance data across 4 tables. The work spans data cleaning, KPI calculation, exploratory data analysis across five dimensions (channel, campaign, region, device, time), and a leads funnel deep dive.

Table	Key Columns	Rows
marketing_dataset	Campaign_ID, Channel, Region, Spend, Revenue, Clicks, Conversions, Device	~3,000
campaigns	Campaign_ID, Budget, Start_Date, End_Date	20
channels	Channel_ID, Channel_Name, Type	5
leads	Lead_ID, Campaign_ID, Channel, Status, Score	19 (sample)

2. Data Cleaning

What Was Fixed

Task	Decision	Why
Column names	Standardized with clean_columns()	Consistency across all tables
Channel & campaign casing	str.strip().str.title()	GOOGLE ADS and google ads are the same thing
Region trailing spaces	str.strip()	Cairo and Cairo were treated as different regions (there is space)
481 missing regions (16%)	Filled with Unknown	No exploitable pattern found across channel, campaign, or device

141 missing lead statuses	Filled with Unknown	Channel signal was only 5-10% variance — too weak to impute
Impressions nulls	Filled with mean	there is no outlier so we use mean
7 rows with revenue = -20	Replaced with 0	All had conversions=0 so how we got revenue !
CPA division by zero	Replace 0 conversions with NaN	Avoids inf values in division
Outlier detection	IQR method on impressions	"IQR detects outliers without being skewed by extreme values."

Key Cleaning Principles Learned

- Weak signal (5-10% group variation) does not justify predictive imputation — Unknown is more honest than false precision
- Choose mean because there is no outlier
- Replace 0 with NaN before dividing — never replace with 1 as it fakes a conversion
- Do not delete rows with bad values if other columns in that row are still valid

3. KPIs Defined & Calculated

KPI	Formula	Finding
CTR	clicks / impressions	~0.1 average — row-level range 0.001 to 0.388
CVR	conversions / clicks	~16% across all channels — flat
CPC	spend / clicks	Facebook lowest ~6.63, Instagram highest ~8.28
CPA	spend / conversions	Facebook cheapest ~97.89, Instagram most expensive ~161.58
ROI	(revenue - spend) / spend	Instagram best ~28x, TikTok worst ~21x — all suspiciously high
ROAS	revenue / spend	Instagram ~29x, mirrors ROI ranking

- **FLAG: ROI values of 21-28x are suspiciously high. Revenue column likely represents total revenue not net profit. Real marketing ROI is 3-10x. Always**

4. Channel Analysis

Channel	ROI	CPA	Total Spend	Verdict
Instagram	28x	161	2.1M	Best ROI, underfunded — scale it
Facebook	25x	98	4M	Cheapest CPA, balanced — maintain
Google Ads	24x	135	4.4M	Most spend, average returns — optimize
Email	21x	129	2.2M	Cheap channel, low ROI — test content
TikTok	21x	122	2.3M	Lowest ROI, high lost rate — review or cut

Lead Status per Channel

- Lost beats or ties Converted in all 5 channels — this is a funnel problem, not a traffic problem
- Google Ads: highest Lost rate ~62% — paying for leads that walk away
- Instagram: Converted and Lost both ~60 — good buyers exist but targeting needs refinement
- Email: most balanced channel — all statuses in tight 31-45 band
- *The business is not losing at the ad level. It is losing after the lead arrives.*

5. Campaign Analysis

Revenue per Campaign

- Ramadan Offer: highest revenue ~49M — clear seasonal winner
- Summer Sale: second ~44M — both seasonal campaigns dominate
- Black Friday, New Launch, Discount Blast: all flat ~20-21M — no differentiation between them

Budget vs Actual Spend

Campaign	Budget	Actual Spend	Variance	Variance %
CMP100 Summer Sale	33,070	7,736	+25,334	+76.6%
CMP101 Black Friday	42,649	19,950	+22,699	+53.2%
CMP102 Black Friday	12,837	9,962	+2,875	+22.4%
CMP103 Black Friday	5,302	35,692	-30,390	-573%
CMP104 Ramadan Offer	42,650	14,710	+27,940	+65.5%
CMP105 Discount Blast	44,874	904	+43,970	+98%

- **FLAG: CMP103 (Black Friday): 5k budget, 35k spent — 573% overspend. Either a data error or zero budget control.**
- **FLAG: CMP105 (Discount Blast): 44k budget, only 904 spent — campaign barely ran.**
- **FLAG: CMP104 (Ramadan Offer): top revenue campaign left 27k unspent — missed scaling opportunity.**
- *The business has a budget execution problem, not a campaign strategy problem. Top performing campaigns are also underspending.*

6. Region Analysis

Region	ROI Rank	CPA Rank	Revenue	Verdict
Tanta	1st	2nd	~27M (3rd)	Efficient, underfunded — scale it
Mansoura	2nd	2nd	~28M (2nd)	Strong across the board — invest more
Cairo	3rd	3rd	~50M (1st)	Biggest market — optimize, do not cut
Unknown	4th	4th	~25M (4th)	Cannot act — fix region tracking first
Alex	5th	5th	~24M (5th)	Worst on everything — needs review

- Cairo generates the most total revenue by a clear margin (~50M vs ~25-28M for everyone else)
- Tanta and Mansoura both run on ~2.5M spend and sit at the top for ROI and CPA — doing more with less
- Alex has the highest CPA and lowest revenue — hardest region to justify continued investment
- *Cairo runs the volume. Tanta and Mansoura run the efficiency. The gap between Cairo spend share and ROI rank is the most actionable finding.*

7. Device Analysis

Device	ROI	CPA	Total Spend	CVR
Mobile	Highest (~9.7x)	Lowest	~5M	~16.8%
Tablet	Middle (~9.0x)	Middle	~5M	~16.5%
Desktop	Lowest (~9.0x)	Highest	~5M	~16.4%

- All three devices perform nearly identically
- Budget split is perfectly equal across all devices
- Mobile technically edges out on ROI and CPA but margin is too small to act on
- **FLAG: In real data you would expect: Mobile = lowest CVR, Desktop = highest CVR. The equal split here is a synthetic data tell.**

8. Time & Seasonality Analysis

Period	Revenue	ROI	Pattern
Q1 (Jan-Mar)	Strong	Strong (~10-12x)	Consistently best quarter across all years
Mid-year (Jul-Sep)	Softer	Dips	Summer underperformance every year
Oct-Nov	Drops	Weakest months	Pre-holiday slump before year-end recovery
December	Peaks	Average	More spend goes in to chase volume

- Feb 2023 hit ~11.8x ROI — best single month in the dataset
- Jun 2025 crashed to ~6.2x ROI — worst month, summer slump at its deepest
- Dec 2024 is the highest revenue point in the entire dataset (~5.7M)
- Oct 2025 is the lowest revenue point (~2.5M) — hard crash before November recovery
- *No overall growth trend. Revenue in Dec 2025 barely higher than Jan 2023. The ROI floor is getting lower each cycle — 2023 floor was ~8x, 2024 was ~7.6x, 2025 dropped to ~6.2x. The business is spending more to generate the same revenue.*
- **FLAG: Campaign timing analysis abandoned — start/end dates in campaigns table were generated independently from transaction dates. Differences of 900+ days make comparison meaningless.**

9. Leads Funnel Analysis

Overall Conversion Rate

- Only 27.1% of leads convert — nearly 3 in 4 leads go nowhere
- This is not a traffic problem. The ads are generating leads. The process after the lead arrives is broken.
- *Before spending more on acquisition, fix the conversion process first.*

Lead Quality by Channel (Score)

- All channels score between 47-50 out of 100 — no meaningful differentiation
- Instagram edges slightly ahead, Facebook slightly behind
- Lead score is not a useful differentiator in this dataset — do not make budget decisions based on score alone

Stuck Leads in Contacted (15.4%)

- 15.4% of all leads were contacted but never moved forward
- These are warm leads — someone already reached out — but they are sitting in limbo
- Either the sales team contacts once and gives up, or the lead goes cold with no follow-up
- *Fixing stuck Contacted leads does not require more ad spend. It requires a better follow-up process.*

Lost Rate per Campaign

- CMP117 loses nearly 4 in 10 leads (~39%) — worst campaign in the dataset
- CMP111 and CMP115 follow closely at ~36%
- CMP101 loses only ~19% — the benchmark every other campaign should study
- The 20-point spread between best and worst means the problem is campaign-specific, not business-wide
- *The business is not uniformly bad at converting leads. It is specifically bad with certain campaigns. That is good news — it is targeted and fixable.*

10. Business Insight Questions

Where is the budget being wasted?

- Google Ads: most spend in the dataset (4.4M) with below-average ROI (24x) and a 62% lead lost rate. Spending the most, not getting the most back.
- CMP103 (Black Friday): 5k budget, 35k actual spend — 573% overspend with no tracking or control.
- CMP105 (Discount Blast): 44k budget, only 904 spent — a campaign that existed on paper but barely ran.
- Alex region: highest CPA, lowest revenue, worst ROI. The numbers do not justify the current budget allocation.
- *Budget waste is not one big problem. It is several small ones — a channel with diminishing returns, a campaign with no controls, and a region with no efficiency.*

Where should we invest more?

- Instagram: highest ROI (28x) on the lowest spend (2.1M). The efficiency is proven. It just needs more budget.
- Tanta region: best ROI, competitive CPA, running on ~2.5M. If efficiency holds at higher spend, it is the clearest regional opportunity.
- Mansoura region: second best ROI, second in revenue, same lean budget as Tanta. Strong across every metric — consistently underinvested.
- Ramadan Offer (CMP104): top revenue campaign left 27k unspent. If it generates the highest revenue while underspending, scaling it is the most obvious move in the entire dataset.
- *The pattern is consistent — the best performers are getting the least money. Shift budget toward Instagram, Tanta, Mansoura, and Ramadan Offer.*

Which campaign should we run again?

- Ramadan Offer: highest total revenue (~49M), second highest ROI, and it underspent its budget — meaning performance was not bought with excessive spend.
- Summer Sale: strong second place in revenue (~44M), consistent performance across channels.
- Both seasonal campaigns outperform every other campaign by a significant margin. The calendar-aligned campaigns win every time.
- *The answer is clear: Ramadan Offer and Summer Sale. The remaining campaigns (Black Friday, New Launch, Discount Blast) are all flat and interchangeable — same audience, same budget, same result.*

Are our leads good quality?

- Conversion rate: 27.1% — nearly 3 in 4 leads do not convert.
- Lead scores: all channels average 47-50 out of 100 — mediocre across the board with no channel standing out.
- 15.4% of leads are stuck in Contacted — warm leads that were reached but never followed up properly.
- Lost rate ranges from 19% to 39% by campaign — CMP117 loses nearly 4 in 10 leads it generates.
- *Lead quality is average at best. The score data says no channel is generating significantly better leads than another. The conversion data says the follow-up process is dropping too many leads that could have been saved.*

What is our best performing region and channel combination?

- Tanta + Instagram: Tanta has the best ROI and efficiency. Instagram has the best ROI among channels. Together they represent the highest-return combination in the dataset.
- Mansoura + Facebook: Mansoura is strong on efficiency and revenue. Facebook has the cheapest CPA (~97). This combination prioritizes volume and cost efficiency over raw ROI.
- Cairo + any channel: Cairo generates the most revenue regardless of channel. The question for Cairo is not which channel — it is how to spend more efficiently in a market that is already performing.
- *Best combination for ROI: Tanta + Instagram. Best combination for volume: Cairo + Facebook. The business should not pick one — it should run both and track them separately.*

11. Data Quality Flags

- **FLAG: Revenue column: likely total revenue not net profit. ROI of 21-28x needs validation before presenting to stakeholders. Real marketing ROI is 3-10x.**
- **FLAG: CTR = 0.1 average: looks flat but row-level variance is real (0.001 to 0.388). Rounding artifact from .round(2).**
- **FLAG: Impressions distribution: ~25k mean per channel with similar std. Confirms synthetic generation.**
- **FLAG: Lead Campaign IDs: CMP115, CMP111 etc in leads table do not exist in campaigns table (CMP100-CMP107). Referential integrity issue.**
- **FLAG: CMP103: 573% overspend — budget entered wrong or spend tracking completely broken for that campaign.**
- **FLAG: Device budget split: perfectly equal thirds across mobile/desktop/tablet. Unrealistic for real data where mobile dominates volume.**

- **FLAG: All channels need ~60 impressions per conversion — identical across channels. In real data this would vary significantly.**
- **FLAG: Campaign start/end dates and actual transaction dates are completely independent. Timing analysis is meaningless on this dataset.**